```
clc;
clear;
close all;
```

PARAMETERS

```
A = 1;                  % Signal amplitude
f = 50;                 % Signal frequency (Hz)
Fs = 1000;              % Sampling frequency (Hz)
N = 1000;               % Number of samples

n = 0:N-1;
x = A * sin(2*pi*f*n/Fs);

B = 8;                  % Main quantizer word length
bits = [4 8 12 16];     % Word lengths for comparison
```

1. 8-BIT QUANTIZATION

```
D = 2*A / 2^B;          % Quantization step size

xr = D * round(x/D);    % Rounding quantization
xt = D * fix(x/D);      % Truncation quantization

er = x - xr;            % Rounding error
et = x - xt;            % Truncation error

% Signal-to-Quantization-Noise Ratio (SQNR)
signalPower = mean(x.^2);

SQr = 10*log10(signalPower / mean(er.^2));
SQt = 10*log10(signalPower / mean(et.^2));
```

2. INPUT / OUTPUT QUANTIZATION

```
xq = xr;                % Quantized input signal
y = x;                  % Ideal output signal

yq = D * round(y/D);    % Quantized output signal
eo = y - yq;            % Output quantization error

SQout = 10*log10(mean(y.^2) / mean(eo.^2));
```

3. WORD-LENGTH COMPARISON

```
step = zeros(size(bits));
sqnrR = zeros(size(bits));
sqnrT = zeros(size(bits));
maxER = zeros(size(bits));
```

```
maxET = zeros(size(bits));

for k = 1:length(bits)

    b = bits(k);
    d = 2*A / 2^b;

    qr = d * round(x/d);
    qt = d * fix(x/d);

    er1 = x - qr;
    et1 = x - qt;

    step(k) = d;

    sqnrR(k) = 10*log10(signalPower / mean(er1.^2));
    sqnrT(k) = 10*log10(signalPower / mean(et1.^2));

    maxER(k) = max(abs(er1));
    maxET(k) = max(abs(et1));

end
```

4. DISPLAY RESULTS

```
fprintf('\n=========================================\n');
fprintf('          QUANTIZATION RESULTS\n');
fprintf('=========================================\n');

fprintf('Amplitude = %.1f\n', A);
fprintf('Frequency = %d Hz\n', f);
fprintf('Sampling Frequency = %d Hz\n', Fs);
fprintf('Number of Samples = %d\n', N);
fprintf('Main Word Length = %d bits\n', B);
fprintf('8-bit Step Size = %.7f\n', D);

fprintf('\nRounding SQNR = %.4f dB\n', SQr);
fprintf('Truncation SQNR = %.4f dB\n', SQt);
fprintf('Output SQNR = %.4f dB\n', SQout);

fprintf('Maximum Rounding Error = %.6f\n', max(abs(er)));
fprintf('Maximum Truncation Error = %.6f\n', max(abs(et)));
```

```
=========================================
          QUANTIZATION RESULTS
=========================================
Amplitude = 1.0
Frequency = 50 Hz
Sampling Frequency = 1000 Hz
Number of Samples = 1000
Main Word Length = 8 bits
8-bit Step Size = 0.0078125
```

```
Rounding SQNR = 48.9339 dB
Truncation SQNR = 45.2931 dB
Output SQNR = 48.9339 dB
Maximum Rounding Error = 0.003483
Maximum Truncation Error = 0.005744
```

5. FIRST 10 SAMPLES TABLE

```
T1 = table(n(1:10)', x(1:10)', xr(1:10)', ...
    er(1:10)', xt(1:10)', et(1:10)', ...
    'VariableNames', {'n', 'Original', 'Rounded', ...
    'RoundError', 'Truncated', 'TruncError'});

fprintf('\n=========================================\n');
fprintf('          FIRST 10 SAMPLES\n');
fprintf('=========================================\n');

disp(T1);
```

```
=========================================
          FIRST 10 SAMPLES
=========================================
    n    Original    Rounded    RoundError    Truncated    TruncError
    _    ________    _______    __________    _________    __________

    0           0          0             0            0             0
    1     0.30902     0.3125     -0.003483      0.30469     0.0043295
    2     0.58779    0.58594     0.0018478      0.58594     0.0018478
    3     0.80902     0.8125     -0.003483      0.80469     0.0043295
    4     0.95106    0.95312    -0.0020685      0.94531      0.005744
    5           1          1             0            1             0
    6     0.95106    0.95312    -0.0020685      0.94531      0.005744
    7     0.80902     0.8125     -0.003483      0.80469     0.0043295
    8     0.58779    0.58594     0.0018478      0.58594     0.0018478
    9     0.30902     0.3125     -0.003483      0.30469     0.0043295
```

6. WORD-LENGTH COMPARISON TABLE

```
T2 = table(bits', step', sqnrR', sqnrT', maxER', maxET', ...
    'VariableNames', {'Bits', 'StepSize', 'SQNR_Round_dB', ...
    'SQNR_Trunc_dB', 'MaxError_Round', 'MaxError_Trunc'});

fprintf('\n=========================================\n');
fprintf('       WORD-LENGTH COMPARISON\n');
fprintf('=========================================\n');

disp(T2);
```

```
=========================================
```

```
       WORD-LENGTH COMPARISON
=========================================
    Bits      StepSize      SQNR_Round_dB     SQNR_Trunc_dB     MaxError_Round
MaxError_Trunc
    ____    __________    _______________    _______________    _______________
_____________

      4           0.125       23.667            20.871
0.059017           0.087785
      8        0.0078125      48.934            45.293
0.003483           0.005744
     12      0.00048828        78.811            65.891
0.00011536         0.00042324
     16      3.0518e-05        99.697            91.599
1.382e-05          2.6516e-05
```

7. OUTPUT PARAMETER TABLE

```
Parameter = {
    'Quantized signal'
    'Quantization error'
    'SQNR'
    'Word-length comparison'
};

Observation = {
    sprintf('8-bit rounding, Delta = %.7f', D)
    sprintf('Max rounding error = %.6f', max(abs(er)))
    sprintf('Rounding = %.4f dB; Truncation = %.4f dB', SQr, SQt)
    'SQNR generally increases with word length'
};

Unit = {
    'Amplitude'
    'Amplitude'
    'dB'
    'Bits / dB'
};

T3 = table(Parameter, Observation, Unit);

fprintf('\n=============================================\n');
fprintf('        OUTPUT PARAMETER TABLE\n');
fprintf('=============================================\n');

disp(T3);


=============================================
        OUTPUT PARAMETER TABLE
=============================================
            Parameter
```

```
Observation                          Unit
    ____________________________
_____________________________________________________    ______________

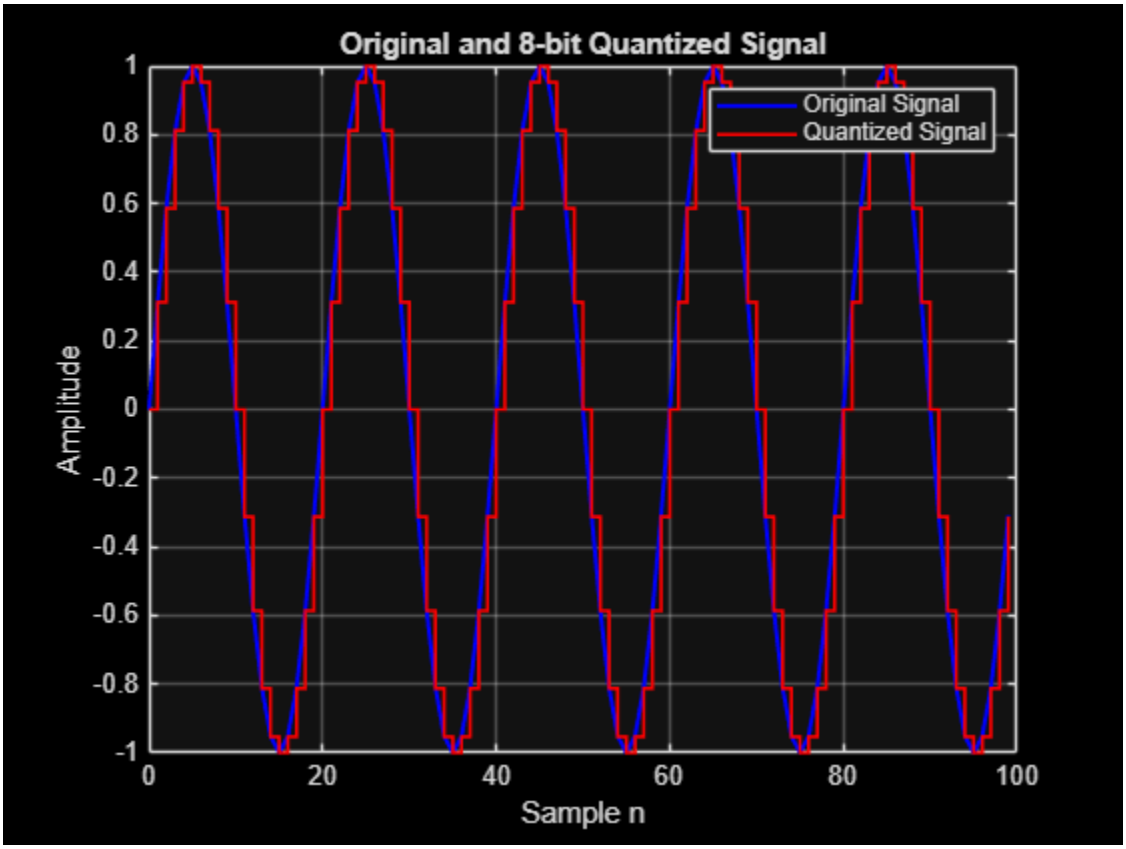
    {'Quantized signal'      }    {'8-bit rounding, Delta =
0.0078125'             }    {'Amplitude'}
    {'Quantization error'    }    {'Max rounding error =
0.003483'                }    {'Amplitude'}
    {'SQNR'                  }    {'Rounding = 48.9339 dB; Truncation =
45.2931 dB'}    {'dB'       }
    {'Word-length comparison'}    {'SQNR generally increases with word
length'     }    {'Bits / dB'}
```

8. GRAPH 1: ORIGINAL AND QUANTIZED SIGNAL

```
figure('Name', 'Original and Quantized Signal');

plot(n(1:100), x(1:100), 'b', 'LineWidth', 1.5);
hold on;
stairs(n(1:100), xr(1:100), 'r', 'LineWidth', 1.2);

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Original and 8-bit Quantized Signal');
legend('Original Signal', 'Quantized Signal');
```

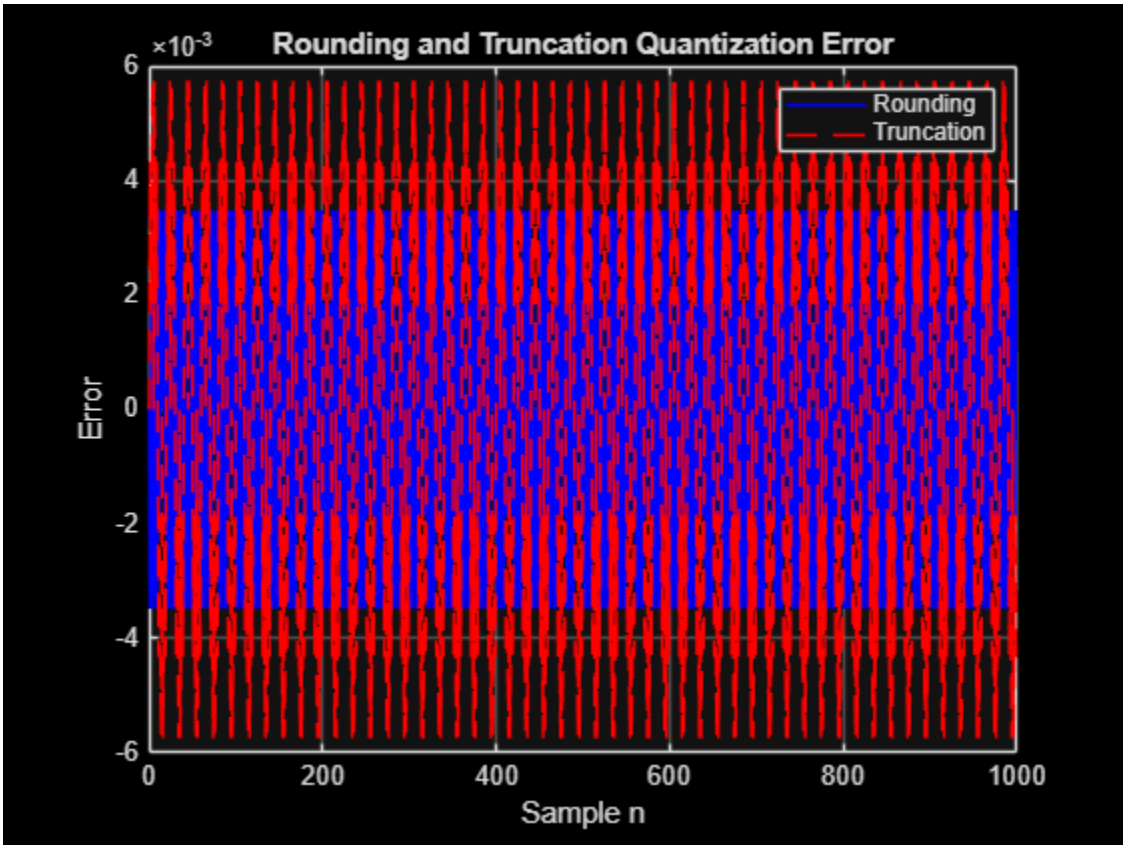


9. GRAPH 2: ROUNDING AND TRUNCATION ERROR

```
figure('Name', 'Quantization Error');

plot(n, er, 'b');
hold on;
plot(n, et, 'r--');

grid on;
xlabel('Sample n');
ylabel('Error');
title('Rounding and Truncation Quantization Error');
legend('Rounding', 'Truncation');
```

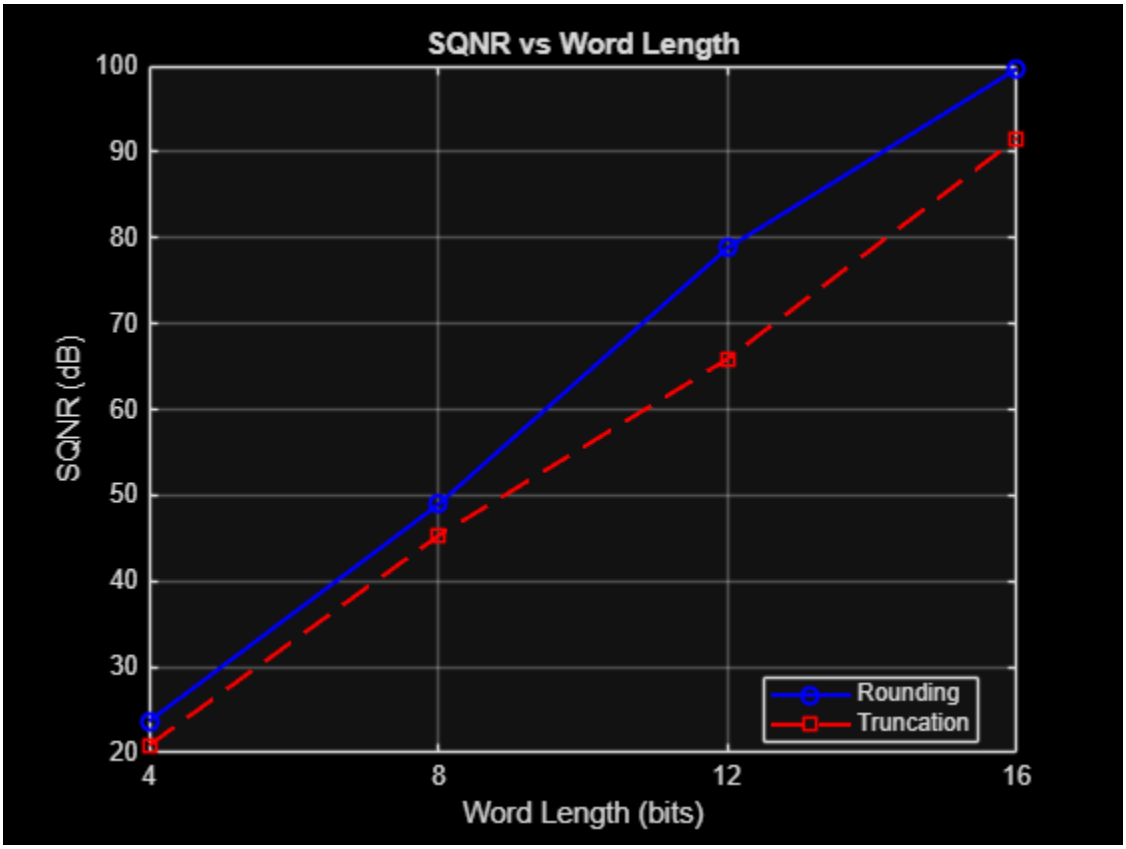


10. GRAPH 3: SQNR VS WORD LENGTH

```
figure('Name', 'SQNR vs Word Length');

plot(bits, sqnrR, 'bo-', 'LineWidth', 1.5);
hold on;
plot(bits, sqnrT, 'rs--', 'LineWidth', 1.5);

grid on;
xlabel('Word Length (bits)');
ylabel('SQNR (dB)');
title('SQNR vs Word Length');
legend('Rounding', 'Truncation', 'Location', 'best');
xticks(bits);
```

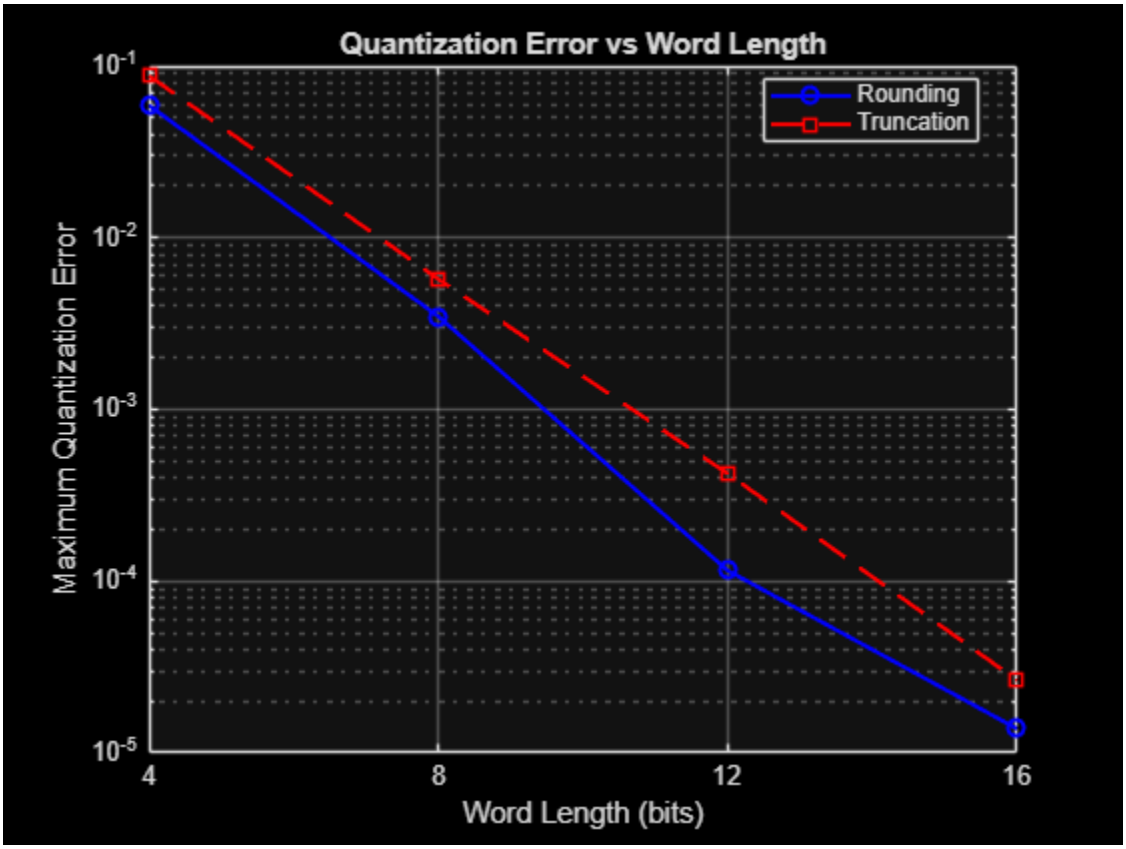


11. GRAPH 4: MAXIMUM ERROR VS WORD LENGTH

```
figure('Name', 'Quantization Error vs Word Length');

semilogy(bits, maxER, 'bo-', 'LineWidth', 1.5);
hold on;
semilogy(bits, maxET, 'rs--', 'LineWidth', 1.5);

grid on;
xlabel('Word Length (bits)');
ylabel('Maximum Quantization Error');
title('Quantization Error vs Word Length');
legend('Rounding', 'Truncation', 'Location', 'best');
xticks(bits);
```



12. FINAL RESULT

```
fprintf('\n==========================================\n');
fprintf('              FINAL RESULT\n');
fprintf('==========================================\n');

fprintf('1. Uniform quantization was performed using rounding and
truncation.\n');
fprintf('2. Quantization error and SQNR were calculated.\n');
fprintf('3. Increasing word length reduces quantization step size.\n');
fprintf('4. Quantization error generally decreases with increasing word
length.\n');
fprintf('5. SQNR generally improves as word length increases.\n');
fprintf('6. Finite word-length effects are verified.\n');
```

```
==========================================
              FINAL RESULT
==========================================
1. Uniform quantization was performed using rounding and truncation.
2. Quantization error and SQNR were calculated.
3. Increasing word length reduces quantization step size.
4. Quantization error generally decreases with increasing word length.
5. SQNR generally improves as word length increases.
6. Finite word-length effects are verified.
```

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